Group Project – Forecasting Incoming Raw Material Deliveries for Hydro ASA

By: Group 72 -
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Agenda

- The Team
- Project Goal
- Problem Definition
- Evaluation Metric
- Our Strategy
- Important Insights From the EDA
- Reproducibility
- Modeling
- Why our Method Worked
- Final Results

The Team

Thomas T.D. Nguyen

5. MSc Economics & Business administration,

Specialization in Business Analytics

+

3. BSc Digital Business Development

Magnus Løwer

4. MSc Computer Science

Project Goal

- **Collaboration:**
  - Append Consulting AS x Hydro ASA
- **Goal:**
  - Use data science/AI to optimize operations with accurate, conservative forecasts of raw material deliveries.
- **Why important?**
  - Avoid shortages in smelting; better planning reduces risks
- **Our role:**
  - Develop ML model

Problem Definition

- Forecast how much **raw material (kg)** will arrive at Hydro's plants **cumulatively** from **January 1st 2025 → any end date up to May 31st 2025**, for each **rm_id**.

- Datasets:
  - Required - receivals.csv (timestamps, net_weight);
  - Recommended - purchase_orders.csv (orders, delivery dates);
  - Optional - materials.csv, transportation.csv.
- Important: Forecasts must be conservative (because Evaluation metrics will penalize overestimation more).

Evaluation Metric

• Quantile Error at 0.2 (Asymmetric Loss): Penalizes overestimation 4x more than underestimation.
• Formula: QuantileLoss0.2(Fi, Ai) = max(0.2*(Ai - Fi), 0.8*(Fi - Ai))
• Overall: Average across all materials (N rm_ids) over h-day windows.
• Why? Underestimation is safer (use on-hand stock); overestimation risks incomplete smelts.

Quantile Error at 0.2 (Asymmetric Loss)

Let there be N raw materials indexed by $i = 1, \ldots, N$. Over a forecasting window of h days, define

$$A_i = \sum_{t=1}^{h} y_{i,T+t}, \qquad F_i = \sum_{t=1}^{h} \hat{y}_{i,T+t},$$

as the actual and forecasted total deliveries for material i, respectively.

To evaluate performance, we compute the quantile loss at the 0.2 level:

$$\text{QuantileLoss}_{0.2}(F_i, A_i) = \max\left(0.2 \cdot (A_i - F_i),\ 0.8 \cdot (F_i - A_i)\right).$$

The overall metric is the average quantile loss across all materials:

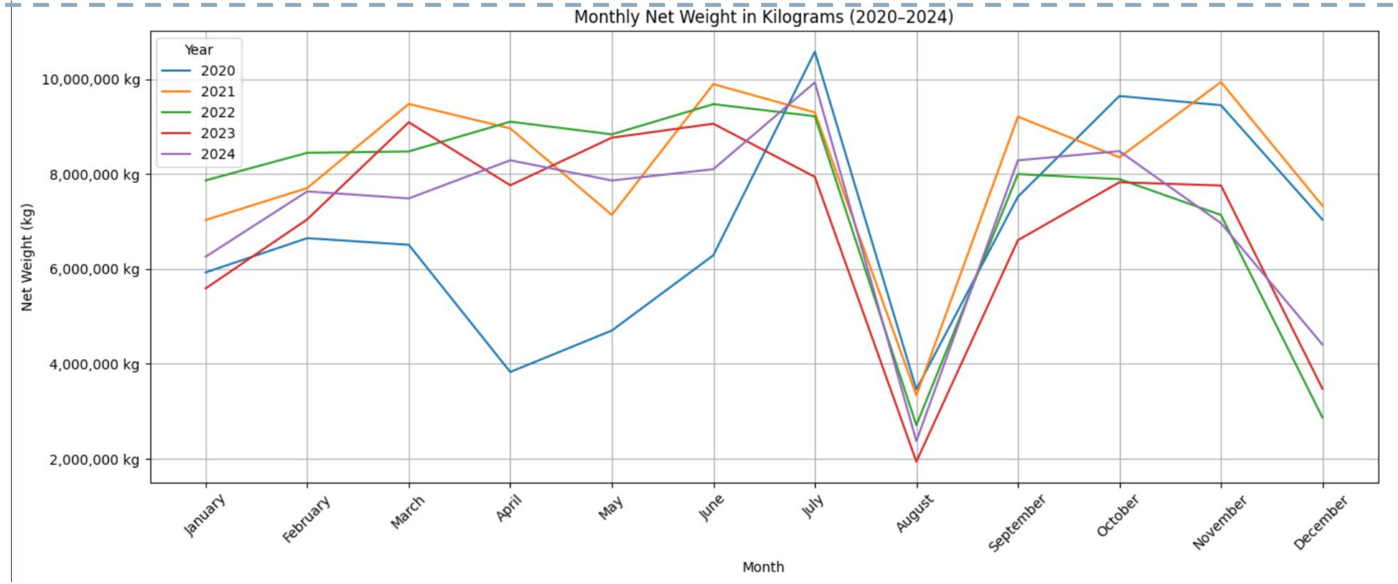
$$\text{QuantileError}_{0.2} = \frac{1}{N} \sum_{i=1}^{N} \text{QuantileLoss}_{0.2}(F_i, A_i).$$

Our strategy

- Exploring Data
- Identify stable, predictable materials
- Build trend-based forecasts on cumulative curves
- Apply conservative slope adjustment
- Avoid overfitting with simple, robust models

Data Exploration (EDA) - Key Insights

- Clear and consistent annual pattern across all five years
- Noticeable drop in August every year (likely holidays/shutdowns)
- High-volume months repeat: June–July and September–November
- Despite year-to-year variation, the shape of the cycle stays stable
- Supports the idea that cumulative deliveries follow a predictable long-term trend



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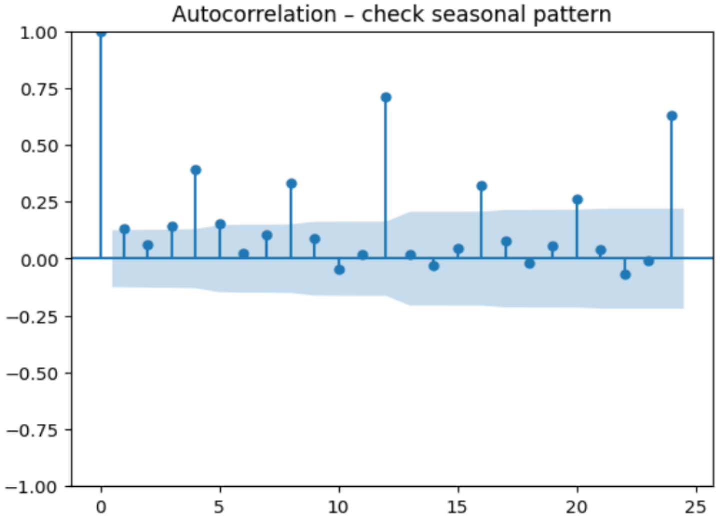
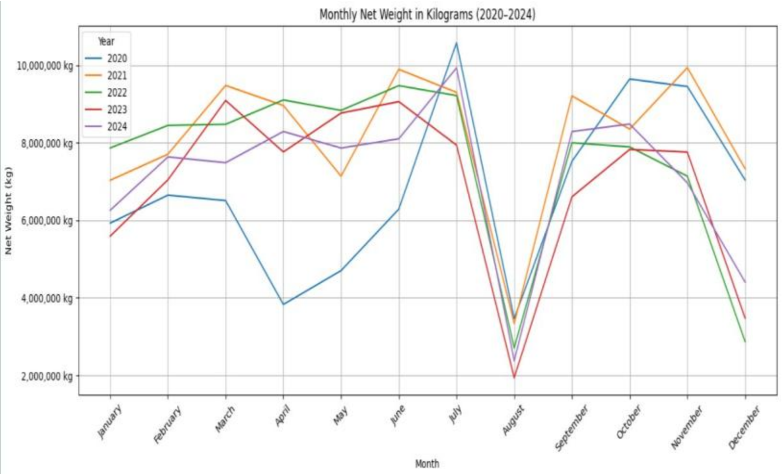
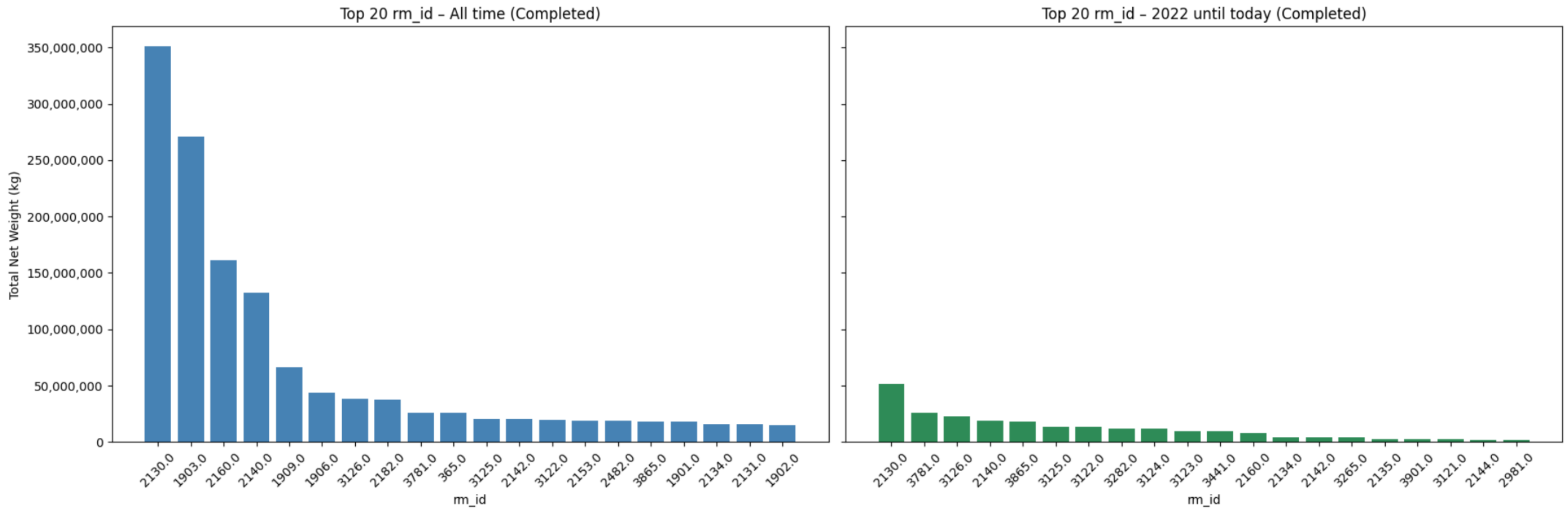




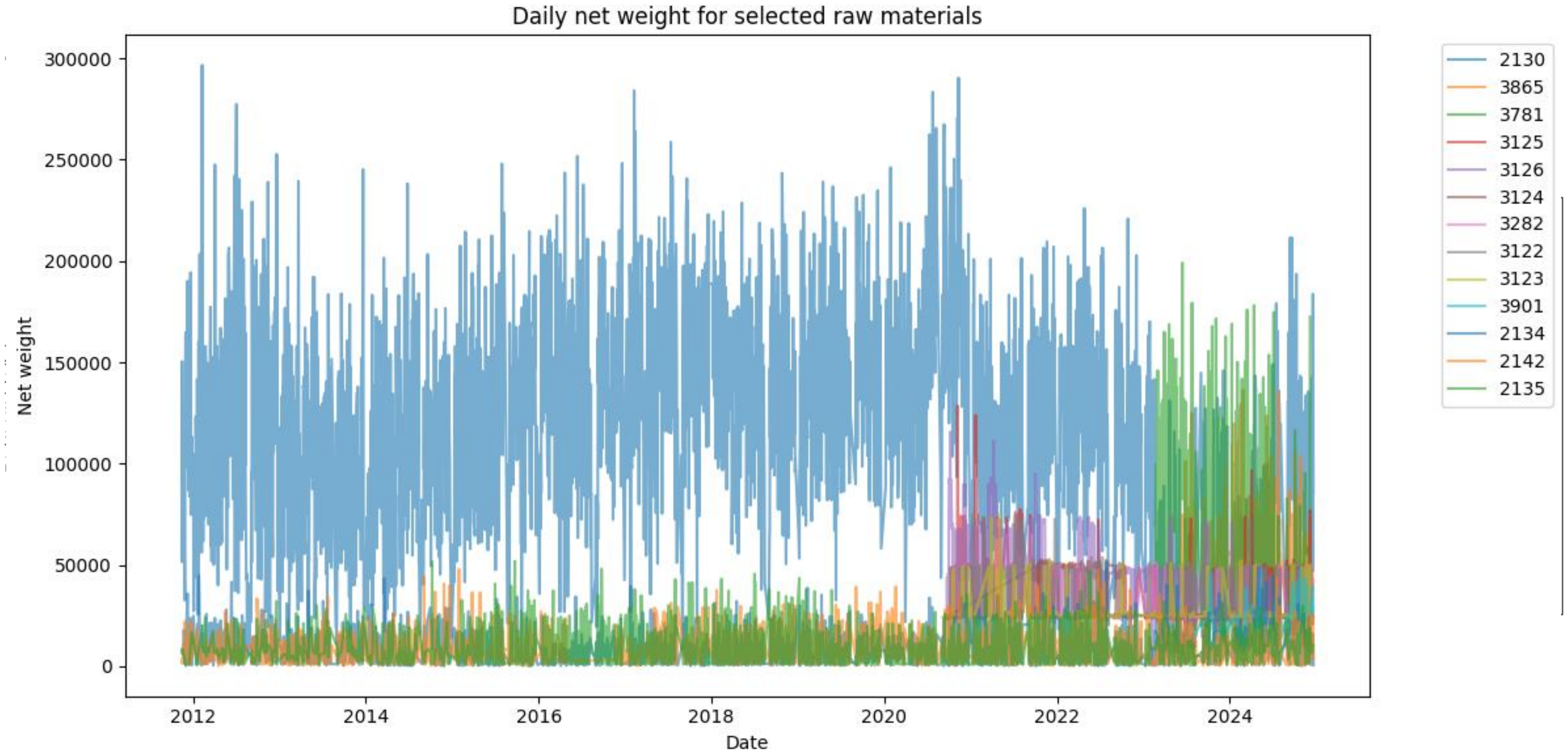
Data Exploration (EDA) - Key Insights

- Checked if every rm_id volume stayed the same
  - Compared every year against 2022 – 2024



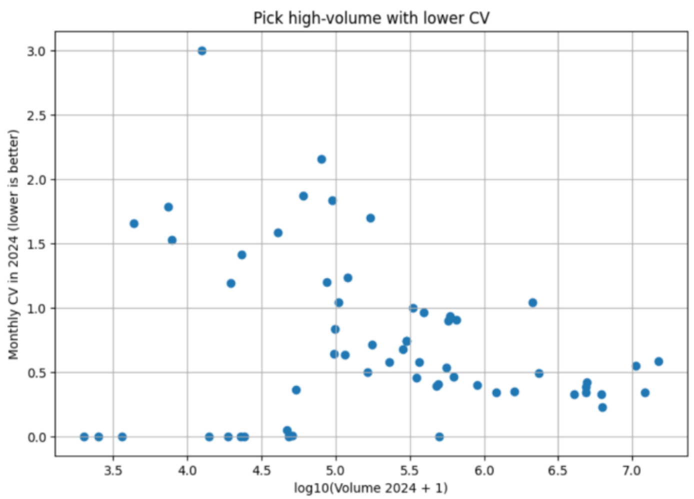
Data Exploration (EDA) - Key Insights

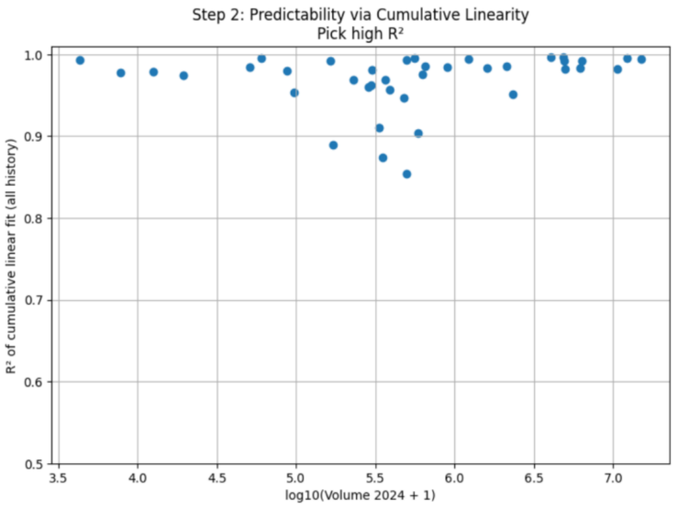
- Daily weight for some raw materials



EDA - Material Selection

- Criteria:
  - Volume (2024),
  - Stability (low Coefficient of Variation),
  - Predictability (high R^2 for linear fit).
- Selected: 20 materials with 12/12 active months





- The Team
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- Problem Definition
- Evaluation Metric
- Our Strategy
- Important Insights

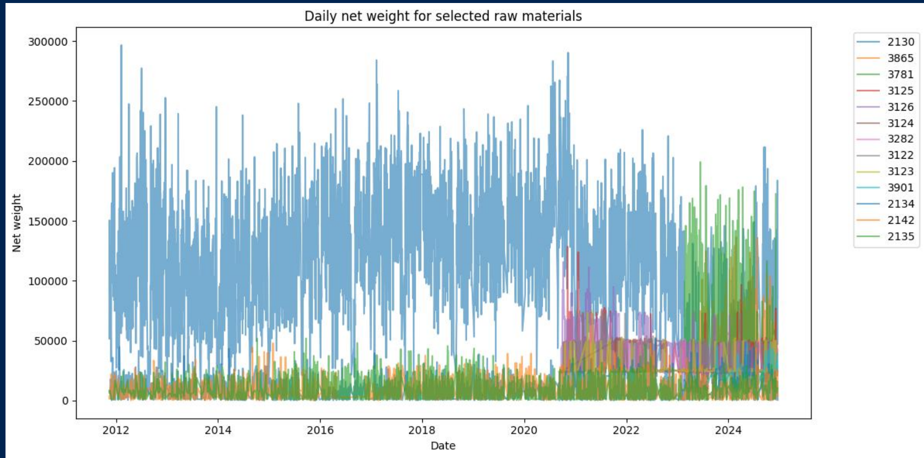
EDA - Material Selection

•Criteria: Volume (2024), Stability (low CV), Predictability (high R^2 for linear fit).
• Selected: 15 (- 2) materials with 12/12 active months

```
RAW_MATERIALS = [2130, 3865, 3781, 3125, 3126, 3124, 3282, 3122, 3123, 3901, 2140, 2134, 2142, 2135, 3265]
MATERIAL_FACTORS = {2130: 0.91, 2134: 0.95, 2135: 0.95, 2140: 0.8, 2142: 0.95, 3122: 0.95, 3123: 0.95, 3124: 0.95, 3125: 0.95, 3126: 0.95, 3265: 0.8, 3282: 0.95, 3781: 0.91, 3865: 0.95, 3901: 0.91}
DEFAULT_FACTOR  = 0.85

Selected materials with diagnostics:
```

rm_id	vol_2024	active_months_2024	cv_2024	cum_r2	top_day_share_2024	factor
2130	15030729.0	12	0.585943	0.994518	0.014080	0.91
3865	12215851.0	12	0.340944	0.994903	0.011152	0.95
3781	10633107.0	12	0.546169	0.981912	0.016751	0.91
3125	6317380.0	12	0.228065	0.991529	0.015275	0.95
3126	6203600.0	12	0.324539	0.982802	0.011400	0.95
3124	4942040.0	11	0.416099	0.982477	0.015160	0.95
3282	4877980.0	12	0.381375	0.992190	0.010369	0.95
3122	4847160.0	12	0.337831	0.996789	0.014928	0.95
3123	4043100.0	12	0.328355	0.996802	0.013411	0.95
3901	2330910.0	12	0.490057	0.950841	0.020713	0.91
2140	2115140.0	7	1.042696	0.985137	0.023639	0.80
2134	1598810.0	12	0.350178	0.983437	0.030526	0.95
2142	1216846.0	12	0.342434	0.994245	0.035880	0.95
2135	902220.0	12	0.397909	0.984403	0.032941	0.95
3265	652080.0	7	0.906006	0.984933	0.076586	0.80



Reproducibility

- Environment: Python 3.10, pip requirements.txt (pandas, scikit-learn, etc.).
- Workflow: Single notebook (Short-notebook_1.ipynb) for end-to-end process.
- Steps: Download folder → Install packages → Run notebook → Get submission.csv.
- Folder: As shown in report (data/kernel, extended; notebooks).

Selected materials with diagnostics:

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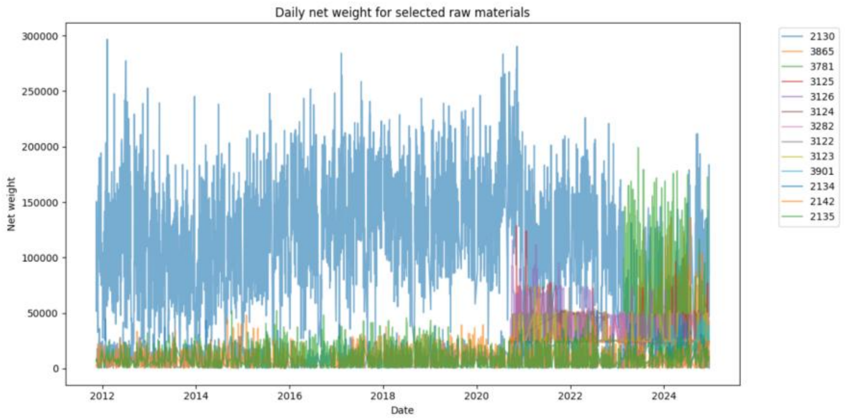
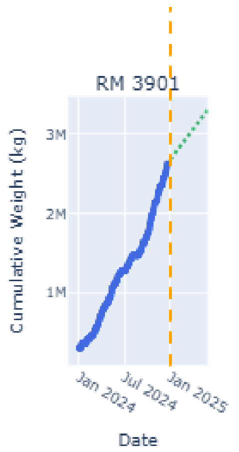
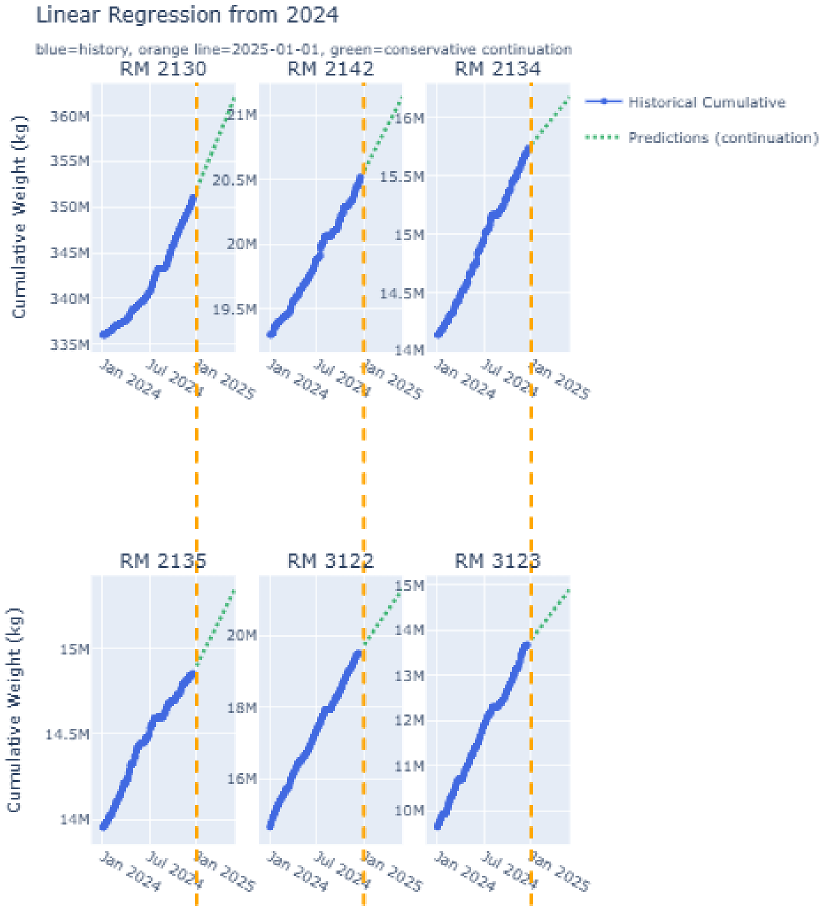


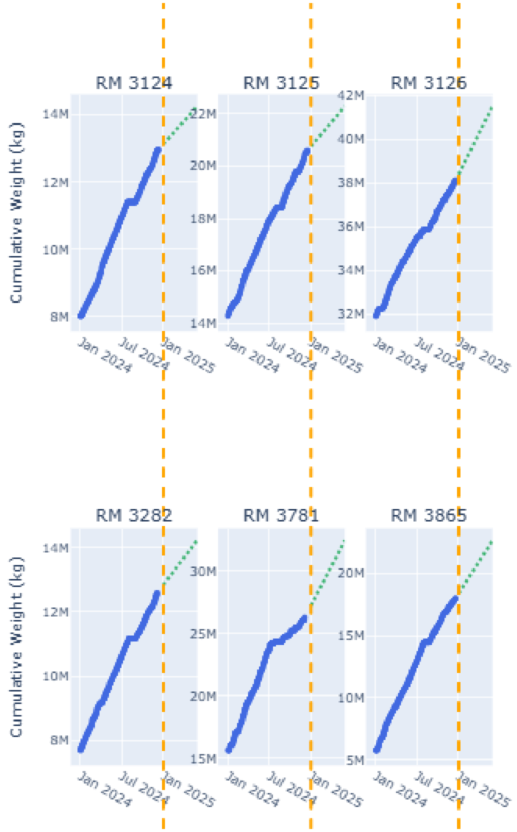
Modeling



RM 3901
Cumulative Weight (kg)
Date



Linear Regression from 2024
blue=history, orange line=2025-01-01, green=conservative continuation
RM 2130
RM 2142
RM 2134
RM 2135
RM 3122
RM 3123
Historical Cumulative
Predictions (continuation)
Cumulative Weight (kg)



RM 3124
RM 3125
RM 3126
RM 3282
RM 3781
RM 3865
Cumulative Weight (kg)

Why our Method Worked

- Focused on creating **accurate, conservative forecasts**
- Did conservative **slope adjustment** for more **reliable** predictions
- Kept the model simple and robust to avoid overfitting
- Ensured the model could generalize well to new data for stable results

Final Results

Final Model Performance
Linear regression on cumulative deliveries per material
Conservative slope adjustment improves Quantile Loss
High R^2 for most selected materials → strong linear trend
Forecasts align well with long-term delivery behavior

Key Takeaways
Simple model performed as well as, or better than, complex alternatives
Avoided overestimation (critical due to asymmetric penalty)
Selected materials showed stability, predictability, and consistent activity

Practical Impact
Reliable short-term forecasts for Hydro's planning
Transparent and interpretable predictions

Thank you for your attention!
Feel free to reach out!

Group 72:

Thomas T.D. Nguyen

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